University of Essex
MSc Artificial Intelligence
Module: IA_PCOM7E January 2026
Unit 1: Collaborative Discussion – Agent-Based Systems

Initial Discussion

The discussion at hand revolves around two topics, each addressed in turn:

1. What are the main factors that led to the development of agent-based systems?

2. What benefits can such systems provide to enterprises and other organizations?

For the first topic, I would assert that agent-based systems have received increasing attention as a response to the need to develop solutions to problems that are, by their nature, unamenable to static or monolithic solutions (Goonatilleke and Hettige, 2022). In other words, when an unchanging, "one size fits all" solution cannot succeed due to the distributed nature of an issue, or the complexity of the issue when encountered at scale, or due to rapidly-changing environmental circumstances, agentic solutions have been developed to fill the gap (Goonatilleke and Hettige, 2022).

For the second topic, agentic solutions offer a variety of advantages. By their nature, they are designed to operate and interact with each other in complex environments with minimal human intervention (Pulikottil et al, 2023). This provides significant opportunities for finding solutions more quickly, at lower cost, and in a more resilient or future-proof manner, as an efficiently-structured agentic system will be intentionally designed to maximize its resilience, and to self-correct, when faced with complex factors in its environment that a static or monolithic solution might not be able to recognize in the first place, or – even if detected – respond to appropriately or effectively (Pulikottil et al, 2023; Tshakwanda, Arzo and Devetsikiotis, 2023).

References

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